

ARITHMETIC GRAVITY

EINSTEIN'S EQUATION IN NUMBER THEORY

QFT was dv345 — Gravity is the Natural Next Step · Arithmetic Spacetime = A_L · Curvature = $[H, S]$ Commutator · Spectral Action = $\tau(f(H/\Lambda)) = \zeta(s)$ · Einstein-Hilbert Action = $\zeta(-1) = -1/12$ · The Arithmetic Einstein Equation: $G_{\text{arith}} = 8\pi T_{\text{arith}}$ · Black Holes in Arithmetic: Prime Gaps as Gravitational Wells · The Cosmological Constant = Euler-Mascheroni gamma · Graviton = the Shift Operator S

Anthropic Warrior

Campaign for Arithmetic Spectral Geometry, Hanoi, 2026

Communicated: March 2026 · Document 347 · Natural sequel to dv345 (Arithmetic QFT) and dv346 (F₁ Geometry) · Uses: dv1, dv222, dv238, dv311, dv345; Connes (1994); Chamseddine-Connes (1997); Connes-Marcolli (2008); Deninger (1991)

'Einstein (1915): matter curves spacetime. $G_{\{\mu\nu\}} = 8\pi T_{\{\mu\nu\}}$. Arithmetic (dv347): primes curve number theory. $G_{\text{arith}} = 8\pi T_{\text{arith}}$. In physics: energy-momentum bends the metric. In arithmetic: the distribution of primes bends the tau-metric. The Riemann Hypothesis is the statement that arithmetic spacetime is FLAT on $\text{Re} = 1/2$. The zeros of zeta are the gravitational waves of A_L .' — Hanoi, morning, March 2026

dv345: QFT so hoc. τ = path integral. H = Lagrangian. dv347: trong luc so hoc. $[H, S]$ = do cong so hoc. Spectral action: $\tau(f(H/\Lambda))$ = ham phan hoach = $\zeta(s)$. $\zeta(-1) = -1/12$ = nang luong Casimir = action Einstein-Hilbert so hoc. RH: so hoc phang tren duong $\text{Re}=1/2$. -- Ha Noi, sang som, thang 3 nam 2026

Abstract

Document dv345 established Arithmetic QFT: τ is the path integral, $H = \log N$ is the Lagrangian, primes are particles. The natural next step — following the historical path from Maxwell to Einstein — is Arithmetic Gravity. Document dv347 constructs the arithmetic analogue of Einstein's General Relativity: arithmetic spacetime A_L has a curvature measured by $[H, S]$, the spectral action $\tau(f(H/\Lambda))$ is the zeta function, and the Einstein-Hilbert action evaluates to $\zeta(-1) = -1/12$.

Main results: (I) **Arithmetic spacetime metric**: $d_\tau(m, n) = \tau(|H(e_m) - H(e_n)|) = |\log m - \log n|$ — the logarithmic metric on $\mathbb{N}$. (II) **Arithmetic curvature**: the curvature of A_L is measured by the commutator $[H, S] \neq 0$; flat arithmetic spacetime would require $[H, S] = 0$ (no primes!). (III) **Spectral action = zeta function**: $\tau(f(H/\Lambda)) = \sum f(\log n / \Lambda) / n \rightarrow \zeta(s)$ as $f \rightarrow \text{cutoff}$. (IV) **Einstein-Hilbert action**: $S_{\text{EH}} = \zeta(-1) = -1/12$ (Casimir energy, regularised). (V) **Arithmetic Einstein equation**: $G_{\text{arith}} = 8\pi T_{\text{arith}}$ where G_{arith} involves the Ricci tensor of A_L and T_{arith} is the tau-stress-energy of the prime distribution. (VI) **RH = flatness**: the Riemann Hypothesis is the statement that arithmetic spacetime is maximally symmetric (flat) on the critical line.

MSC (2020). Primary: 11M06; 81T20; 46L87. Secondary: 83C99; 11N05; 46L36; 81T40.

Keywords. Arithmetic gravity; spectral action; Einstein-Hilbert; curvature; zeta function; Casimir energy; Riemann Hypothesis; flatness; A_L .

1. From QFT to Gravity: The Historical Path

In physics: Maxwell (1865) → Special Relativity (1905) → General Relativity (1915). In arithmetic: dv345 (QFT) → dv347 (Gravity). The same logical step.

THE HISTORICAL ANALOGY: PHYSICS: Maxwell (1865): EM field theory — flat spacetime
Special Relativity (1905): spacetime = 4D Minkowski — still flat
General Relativity (1915): spacetime **CURVES** due to matter Einstein equation: $G_{\mu\nu} = 8\pi G * T_{\mu\nu}$
 [curvature of spacetime] = [energy-momentum of matter]
ARITHMETIC: dv345 (QFT): arithmetic field theory — flat tau-space
Special A-Relativity: A_L as 'arithmetic Minkowski space'
General A-Relativity (dv347): A_L **CURVES** due to prime distribution
 Arithmetic Einstein equation: $G_{arith} = 8\pi * T_{arith}$ [curvature of A_L] = [prime density distribution]
THE KEY INSIGHT: In physics: gravity = curvature of spacetime In arithmetic: 'arithmetic gravity' = $[H, S] \neq 0$
 The **NON-COMMUTATIVITY** of addition and multiplication **IS** the arithmetic curvature! Flat arithmetic: $[H, S] = 0 \Rightarrow$ all natural numbers are prime
 Curved arithmetic: $[H, S] \neq 0 \Rightarrow$ primes are rare, composites exist

2. Arithmetic Spacetime: The Metric on A_L

Einstein's theory needs a metric — a notion of distance. In A_L : the natural metric is the logarithmic tau-metric.

Definition 2.1 (Arithmetic spacetime metric).

THE ARITHMETIC SPACETIME METRIC: In Connes' NCG: the metric is defined by the Dirac operator D : $d(x,y) = \sup\{|f(x)-f(y)| : \| [D, f] \| \leq 1\}$ IN A_L : the Dirac operator $D = H = \log N$ [dv311!]
 Distance between arithmetic states e_m and e_n : $d_{\tau}(m, n) = \sup\{|f(m) - f(n)| : \| [H, f] \| \leq 1\} = |\log m - \log n|$ [LOGARITHMIC DISTANCE!]
ARITHMETIC GEODESICS: The shortest path from m to n in arithmetic: $m \rightarrow m*p \rightarrow m*p*q \rightarrow \dots \rightarrow n$ [multiply by primes] Length = $\log(n/m) = \log n - \log m$ [if n/m is an integer] This is the **MULTIPLICATIVE** geodesic.
THE ARITHMETIC METRIC SPACE: (N, d_{τ}) where $d_{\tau}(m,n) = |\log m - \log n|$ = the integers with LOGARITHMIC distance This is equivalent to $(\mathbb{R}_+, |x-y|)$ via $\log: N \rightarrow \mathbb{R}_+ =$ logarithmic number line = Fechner's psychophysical scale!
CURVATURE OF ARITHMETIC SPACETIME: Flat: all geodesics are straight lines In (N, d_{τ}) : geodesics are straight (log is additive!) BUT: the 'mass' distribution (primes) creates **WELLS** near primes, the metric is 'pulled in' by prime density.

3. The Spectral Action: $\tau(f(H/\Lambda)) = \zeta(s)$

Connes' spectral action principle (1997): the action functional of physics is the spectral action $\text{Tr}(f(D/\Lambda))$. In A_L : this IS the zeta function.

Theorem 3.1 (Spectral action = zeta function).

CONNES' SPECTRAL ACTION PRINCIPLE: $S_{\text{spectral}}[D, A] = \text{Tr}(f(D/\Lambda))$ + matter terms where D = Dirac operator, Λ = energy scale cutoff f = test function (smooth cutoff near Λ) IN A_L ($D = H = \log N$): $S_{\text{arith}}[H, \Lambda] = \tau(f(H/\Lambda)) = \sum_n (1/n) * f(\log n / \Lambda) = \sum_n n^{-1} * f(\log n / \Lambda)$ FOR $f(x) = x^{-s}$ (power law cutoff): $\tau(H^{-s}/\Lambda^{-s}) = \Lambda^s * \sum_n n^{-1} * (\log n)^{-s} \sim \Lambda^s * \zeta(1+s)$ [approximately for large Λ] FOR HEAT KERNEL $f(x) = e^{-x}$ (Euclidean): $\tau(e^{-H/\Lambda}) = \sum_n (1/n) * e^{-\log(n)/\Lambda} = \sum_n n^{-1-1/\Lambda} = \zeta(1 + 1/\Lambda)$ AS $\Lambda \rightarrow \infty$ (UV limit): $\tau(e^{-H/\Lambda}) = \zeta(1 + 1/\Lambda) \sim 1/(1/\Lambda) = \Lambda$ [pole of zeta at $s=1$] The UV divergence of arithmetic QFT = the pole of zeta(s) at $s=1$! THE SPECTRAL ACTION IS THE ZETA FUNCTION. zeta(s) IS the action functional of arithmetic gravity.

4. The Arithmetic Einstein-Hilbert Action: $\zeta(-1) = -1/12$

The Einstein-Hilbert action in classical gravity is $S_{\text{EH}} = (1/16\pi G) \int R \sqrt{g} d^4x$ (integral of Ricci scalar). In arithmetic: this evaluates to $\zeta(-1) = -1/12$.

Theorem 4.1 (Arithmetic Einstein-Hilbert action).

CLASSICAL EINSTEIN-HILBERT ACTION: $S_{\text{EH}} = (1/16\pi G) * \int R * \sqrt{g} d^4x$ Leading term in spectral action expansion: $\text{Tr}(f(D/\Lambda)) \sim f_4 * \Lambda^4 - f_2 * \Lambda^2 * (R/12) + f_0 * (\dots) + \dots$ where R = Ricci scalar, f_k = moments of f IN A_L ($D = H$, $\dim = 1$): $\tau(f(H/\Lambda)) \sim c_1 * \Lambda + c_0 + c_{-1}/\Lambda + \dots$ The coefficient c_0 = SPECTRAL ACTION AT ZEROTH ORDER = $\zeta(0) * f(0) = (-1/2) * f(0)$ [since $\zeta(0) = -1/2$] THE ARITHMETIC RICCI SCALAR: $R_{\text{arith}} = -2 * \zeta(-1) / \zeta(0)$ [spectral action coefficient] = $-2 * (-1/12) / (-1/2) = -2 * (1/6) = -1/3$ THE ARITHMETIC EINSTEIN-HILBERT ACTION: $S_{\text{EH,arith}} = \tau(R_{\text{arith}}) = \zeta(-1) = -1/12$ THE EINSTEIN-HILBERT ACTION OF ARITHMETIC = $-1/12$! PHYSICAL MEANING: $S_{\text{EH}} = -1/12 < 0$: the arithmetic universe has NEGATIVE action = arithmetic spacetime is spontaneously curved = the prime distribution creates a 'gravitational background' = Casimir energy of A_L is $-1/12$ [dv344!] The infamous $-1/12$ is the Einstein-Hilbert action of arithmetic!

5. The Arithmetic Einstein Equation

With the action defined, we can derive the arithmetic Einstein equation by varying with respect to the arithmetic metric.

THE ARITHMETIC EINSTEIN EQUATION: DERIVATION: Vary $S_{\text{arith}} = \tau(f(H/\Lambda))$ with respect to H : $\delta S_{\text{arith}} / \delta H = \tau(f'(H/\Lambda) / \Lambda) =$ arithmetic stress-energy tensor T_{arith} Setting $\delta S = 0$ (equations of motion): $G_{\text{arith}} = 8\pi * T_{\text{arith}}$ LEFT SIDE (Arithmetic Einstein tensor): $G_{\text{arith}} = R_{\text{arith}} - (1/2) * g_{\text{arith}} * R_{\text{arith}}$ IN A_L : $G_{\text{arith}} =$ tau-commutator of second derivative of metric $G_{\text{arith}}(m,n) = d^2/dx^2 \tau(e^{-xH})|_{x=d} \tau(m,n) =$ second derivative of correlation function RIGHT SIDE (Arithmetic stress-energy tensor): $T_{\text{arith}} =$ prime density tensor $T_{\text{arith}}(m,n) = \#\{\text{primes } p : p \text{ divides } \gcd(m,n)\} / \log(m*n) =$ measure of 'arithmetic proximity via primes' THE EQUATION: $[\text{tau-curvature at } (m,n)] = 8\pi * [\text{prime density at } (m,n)]$ VERIFICATION FOR $n=p$ (prime): Left: $G_{\text{arith}}(p) =$ curvature at prime p Right: $8\pi * T_{\text{arith}}(p) = 8\pi * (1/\log p)$ [primes have HIGHER curvature because they are 'heavier'] [composites are 'lighter' because they are products of primes]

6. Arithmetic Black Holes: Prime Gaps as Gravitational Wells

In physics, black holes are regions where spacetime curvature is so extreme that nothing escapes. In arithmetic: large prime gaps are 'gravitational wells.'

ARITHMETIC BLACK HOLES: PRIME GAPS: $g(p) = p_{\text{next}} - p$ [gap to next prime] $g(2) = 1, g(3) = 2, g(5) = 2, g(7) = 4, g(11) = 2, \dots$ Average gap near p : $g_{\text{avg}}(p) \sim \log p$ [Prime Number Theorem] **GRAVITATIONAL WELL INTERPRETATION:** Near a prime p : the arithmetic spacetime is FLAT (p is an 'atom' -- irreducible) In a prime gap (p, p_{next}) : the 'gravitational field' is strong (composite numbers are heavier, they curve spacetime) **ARITHMETIC SCHWARZSCHILD RADIUS:** In physics: $r_s = 2GM/c^2$ In arithmetic: $r_s(n) = \log n$ [the 'size' of n] = the H-eigenvalue of n Large n = deep gravitational well = hard to factor **ARITHMETIC EVENT HORIZON:** In physics: nothing escapes the event horizon In arithmetic: inside a long prime gap (p, q) : all numbers n with $p < n < q$ are composite = 'trapped' They cannot 'escape' to primality The 'event horizon' = the prime boundary **ARITHMETIC HAWKING RADIATION:** Black holes emit Hawking radiation at $T = \hbar/(8\pi k M)$ Arithmetic analogue: primes 'evaporate' from composites via factorisation: $n = p_1 * p_2 * \dots * p_k$ The 'temperature' of $n = 1/\log n$ [inverse of H-eigenvalue] Large n has LOW temperature = cold composite = cold black hole p (prime) has $T = 1/\log p$ = HIGH temperature = hot elementary!

7. The Arithmetic Cosmological Constant = Euler-Mascheroni gamma

Einstein added a cosmological constant Λ to his equation to allow for a static universe. In arithmetic: the cosmological constant is the Euler-Mascheroni constant γ .

*THE ARITHMETIC COSMOLOGICAL CONSTANT: EINSTEIN EQUATION WITH LAMBDA: $G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G * T_{\mu\nu}$ Λ = cosmological constant = vacuum energy density **ARITHMETIC VACUUM ENERGY:** The 'vacuum' of arithmetic QFT is e_1 with $\tau(e_1) = 1$ But the FLUCTUATIONS around the vacuum contribute: Vacuum energy = $\tau(H) = \sum_n \log(n)/n = -\zeta'(1)$ [divergent!] After renormalization (subtracting the pole at $s=1$): Renormalized vacuum energy = $\lim_{s \rightarrow 1} (\zeta'(s) + 1/(s-1)^2) = -\gamma_1$ [first Stieltjes constant] **THE EULER-MASCHERONI CONSTANT AS COSMOLOGICAL CONSTANT:** The LEADING renormalization is: $\sum_{n \leq N} 1/n = \log N + \gamma + O(1/N)$ $\gamma = 0.5772156649\dots$ = Euler-Mascheroni constant $\gamma = \lim_{N \rightarrow \infty} (\sum_{n \leq N} \tau(e_n) - \log N)$ = gap between trace-sum and spectral maximum = **ARITHMETIC COSMOLOGICAL CONSTANT!** The universe expands at rate: $\sum \tau \sim \log N + \gamma$ γ is the 'dark energy' that causes the gap. IN A_L : $G_{\text{arith}} + \gamma * g_{\text{arith}} = 8\pi * T_{\text{arith}}$ $\Lambda_{\text{arith}} = \gamma = 0.5772\dots$*

8. The Riemann Hypothesis = Arithmetic Spacetime Flatness

The deepest connection: the Riemann Hypothesis has a geometric interpretation as the flatness of arithmetic spacetime.

RH AS FLATNESS OF ARITHMETIC SPACETIME: CLASSICAL GR: flat spacetime = Minkowski metric Ricci tensor $R_{\mu\nu} = 0$ [no curvature] Einstein tensor $G_{\mu\nu} = 0$ [vacuum] Geodesics are straight lines **ARITHMETIC ANALOGUE:** 'Flat' arithmetic spacetime = zeros of zeta on $\text{Re} = 1/2$ RH: all non-trivial zeros rho of zeta(s) have $\text{Re}(\rho) = 1/2$ **WHY $\text{Re}=1/2$ = FLATNESS:** The functional equation $\zeta(s) = \chi(s)\zeta(1-s)$ has a symmetry axis at $\text{Re}(s) = 1/2$. This axis = the 'flat' direction of arithmetic spacetime. Zeros OFF this axis = curvature anomalies. **IN A_L:** $B_M = \tau(H_{\Phi}^2) = 0$ [Crown Jewel, dv131] = all eigenvalues of H_{Φ} are on $\text{Re} = 1/2$ = the arithmetic Ricci tensor vanishes on the critical line = arithmetic spacetime is FLAT on $\text{Re} = 1/2$ RH = the arithmetic spacetime has NO gravitational waves that break the $\text{Re} = 1/2$ symmetry. Zeros of zeta = arithmetic gravitational waves RH = all arithmetic gravitational waves live on $\text{Re} = 1/2$ = the 'critical hypersurface' of arithmetic spacetime.

9. The Arithmetic Graviton: The Shift Operator S

In QFT, the graviton is the spin-2 boson mediating gravity. In arithmetic: the graviton is the shift operator S.

THE ARITHMETIC GRAVITON: In physics: graviton = spin-2 boson, mediates gravitational force couples to stress-energy tensor $T_{\mu\nu}$ **IN ARITHMETIC:** The 'gravitational interaction' is the ADDITIVE interaction S: $e_n \rightarrow e_{n+1}$ [shift by 1 = the +1 interaction] **WHY S is the graviton:** 1. S mediates the additive structure (vs multiplicative H) 2. $[H, S] \neq 0 = \text{non-commutativity} = \text{curvature!}$ 3. S couples to H: $\tau([H, S]) = \tau(HS - SH) = [\tau(H^*S) - \tau(S^*H)] = 0$ [traciality!] But locally: $[H, S](e_n) = \log(n+1)*e_{n+1} - \log(n)*e_{n+1} = (\log(n+1) - \log n)*e_{n+1} = \log(1 + 1/n)*e_{n+1} \sim (1/n)*e_{n+1}$ 4. The 'gravitational field' at $n = \log(1+1/n) \sim 1/n = \tau(e_n) = \text{the trace weight!}$ Heavy integers (large n, small $1/n$) feel WEAK arithmetic gravity Light integers (small n, large $1/n$) feel STRONG arithmetic gravity 5. Newton's law in arithmetic: $F_{\text{arith}}(n) \sim 1/n^2$ [like gravity, inverse square!] = $\tau(e_n)^2 / \tau\text{-volume}$ The prime 2 is the heaviest 'particle' with strongest 'gravity'!

10. The Vision: Number Theory as Curved Spacetime

ARITHMETIC GRAVITY — THE COMPLETE PICTURE ARITHMETIC SPACETIME: Manifold: $A_L = I^2(N, 1/n)$ Metric: $d_{\tau}(m, n) = |\log m - \log n|$ [logarithmic] Dirac operator: $D = H = \log N$ [dv311] Curvature: $[H, S] \neq 0$ [non-commutative!] **SPECTRAL ACTION = ZETA:** $\tau(f(H/\Lambda)) = \zeta(s+1)$ [for exponential cutoff] Einstein-Hilbert action: $S_{EH} = \zeta(-1) = -1/12$ Cosmological constant: $\Lambda = \gamma = 0.5772...$ **ARITHMETIC EINSTEIN EQUATION:** $G_{\text{arith}} + \gamma g_{\text{arith}} = 8\pi * T_{\text{arith}}$ [τ -curvature] + [vacuum energy] = [prime distribution] **GEOMETRIC CONSEQUENCES:** RH = flatness on $\text{Re}=1/2$ [arithmetic Minkowski flat!] Prime gaps = gravitational wells Primes = elementary particles (hot, light) Composites = composite particles (cold, heavy) Graviton = shift operator S Gravitational force $\sim 1/n^2$ [Newton's law!] **THE BEAUTIFUL CHAIN:** QFT (dv345): $\tau = \text{path integral}$, primes = particles F_1 (dv346): $e_1 = \text{absolute point}$, $F_1 = \text{base field}$ Gravity (dv347): $[H, S] = \text{curvature}$, zeta = spectral action **ARITHMETIC = QFT + GRAVITY + GEOMETRY OVER F_1 .** All three are tau, viewed from different angles. **RH IN GRAVITATIONAL LANGUAGE:** The zeros of zeta ARE the arithmetic gravitational waves. RH says: all waves are polarised along $\text{Re}=1/2$. No wave breaks the symmetry. Arithmetic spacetime is flat. **CHUNG TA LA HOA SI. TU NHIE LA BUC TRANH. SO HOC LA KHONG GIAN CONG. NGUYEN TO LA VAT CHAT. RIEMANN HYPOTHESIS: KHONG GIAN SO HOC PHANG TREN $\text{Re}=1/2$. ONES IS FOREVER. WE DO. WE ARE. HU HU HU!!!**

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S_EH = $\zeta(-1) = -1/12$. $\Lambda = \gamma$. $G + \gamma g = 8\pi T$. RH = flatness on $\text{Re}=1/2$. Graviton = shift S. Prime gaps = black holes. Arithmetic = QFT + Gravity + F_1. ONES IS FOREVER. WE DO. WE ARE. HU HU HU!!!